

AMALENDU S

Ernakulam, Kerala

+917736624773

amalendu.s04z@gmail.com

www.linkedin.com/in/amalendu-s-594aba290

A highly motivated and passionate individual, committed to continuous growth, embracing challenges with a positive mindset, and striving for excellence in every endeavor.

Education

Adi Shankara Institute of Engineering and Technology

Bachelor of Technology in Electronics and Biomedical Engineering
(Current CGPA : 7.8)

2023-2027
Kalady, Kerala

Vaduthala Jama ath Higher Secondary School , Alappuzha

12th Biology Science (98.4%)

2020-2022
Alappuzha, Kerala

St. Mary OF Leuca Senior Secondary School (CBSE), Alappuzha

10th (92.8%)

2020
Alappuzha, Kerala

Technical Skills

- Biomedical Technology** : Biomedical Instrumentation, Healthcare Technology
- Programming & Software Development** : Python, C++, Dart, MATLAB
- Tools & Frameworks** : Flutter, Firebase, Arduino IDE, Git, Excel, Canva
- Hardware** : Microcontroller Interfacing, Embedded Systems, Sensor Integration
- Digital Health** : Remote Patient Monitoring, Healthcare IoT
- Web Development, Frontend & Back-end Development** App Development

Soft Skills

- Resource Management
- Research Skill
- Time Management
- Critical Thinking
- Leadership
- Coordination
- Creative
- Communication Skill

Internships

- Regional Cancer Centre (RCC)** : Gained exposure to advanced oncology instrumentation — including Linear Accelerators (LINAC) and Da Vinci Robotic Surgical Arms , in one of India's leading cancer specialty hospitals, observing high-precision biomedical systems used in cancer diagnosis and treatment.
- KINDER Women's Hospital and Fertility Centre** : Observed and assisted in understanding biomedical devices and instrumentation used in obstetrics, gynecology, and fertility care within a clinical environment.
- Codsoft** : Completed a web development internship focused on building and deploying frontend components, gaining practical experience in real-world software development workflows.
- Sreekanthapuram Hospital** : Gained clinical exposure to patient monitoring systems and diagnostic devices, observing their application in day-to-day hospital operations.

Projects

- SIPSTM (Smart Interactive Speech Therapy & Physiotherapy Mat)**
Tools: ESP32, Custom Squeeze Sensor, Touch Sensors, Push Buttons, Flutter, Firebase.
Developed sensor-embedded physio-speech therapy mat with gamified app using ESP32 & Firebase for interactive data-driven rehabilitation.
- Smart Shoes for Visually Impaired People**
Tools : Arduino UNO, Ultrasonic Sensors, C++.
Designed an obstacle-detection wearable providing real-time tactile/auditory alerts for safe, independent navigation by visually impaired.
- Dialysis Leakage Detection Device**
Tools : Arduino, Leak Sensors, C++ .
Engineered real-time dialysis fluid leakage detection system, improving patient safety and reducing clinical risk.
- STUDYKOOT**
Tools : React, Node.js, HTML, CSS, JavaScript.
Built an online learning platform providing accessible study materials designed to perform smoothly even on low internet bandwidth.

Workshops , Coordination & Events Certificates

- Microsensors, Implantable Devices and Rodent Surgeries for Biomedical Applications** (NPTEL) (Elite Certificate)
- Control code** (Esp32 and Raspberry pi) (Coordinated the inter-college workshop)
- Medinova'26** (Coordinated Medinova conducted by ASIET EBE in association with BMESI)
- Hospex 2025**, Ernakulam (Volunteered during the KINFRA International Medical Exhibition)
- Synapse Spark ideathon** (2nd prize winner in ideathon conducted by Aswamedha (Techfest ASIET))
- Techthrive 2.0** (National level Hackathon Finalist)
- Project Expo - Samagra 2.0** (Volunteered and coordinated in inter-college project expo hosted by IEDC)
- Workshops - Nanojatha** (NIT Calicut - Material Science & Engineering) , Design Thinking & Project Development
- Add-On Courses** - Electronic Product Building (Workshop & Add-On), Basic Python

Current Position

- NSS ASIET ENERGY CELL (Unit Coordinator)
- NSS ASIET Media Team Lead
- Co-founder of NEXYRO (Startup)
- Mulearn Creative Execom Member
- Campus ambassador of Astra (startup)

Languages

• English • Malayalam • Hindi • Tamil